

Project Report

Project Title: A Comparative Study of Machine Learning Models for
Clinical Insights

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1. Introduction

1.1 Project Overview

The Heart Disease Prediction project leverages machine learning to predict the likelihood of heart disease in patients based on various health metrics. Utilizing a dataset from the UCI repository, this project compares different machine learning models, selects the best-performing one, and integrates it into a web application for real-time prediction.

1.2 Purpose

The goal of this project is to create a robust prediction tool that can assist healthcare providers and individuals in assessing heart disease risk. Early prediction can help in taking preventive measures and thus, reduce mortality rates related to heart disease.

2. Literature Survey

2.1 Existing Problem

Heart disease is a leading cause of death globally, with significant risks associated with late diagnosis. Traditional diagnostic approaches are resource-intensive, leading to high costs and reduced accessibility.

2.2 References

[Thunshun W. Liao, 2021] Comparative Study on Heart Disease Prediction Using Feature Selection Techniques on Classification Algorithms | [Link](#)

2.3 Problem Statement Definition

Develop a machine learning-based predictive model to estimate heart disease risk based on user-provided health data, utilizing models optimized for accuracy, precision, and recall.

3. Requirement Analysis

3.1 Functional Requirements

- The system should allow users to input personal health metrics.
- The model should process data and return a prediction on heart disease likelihood.

- The application must provide visualization of results, such as a pie chart showing risk distribution.

3.2 Non-Functional Requirements

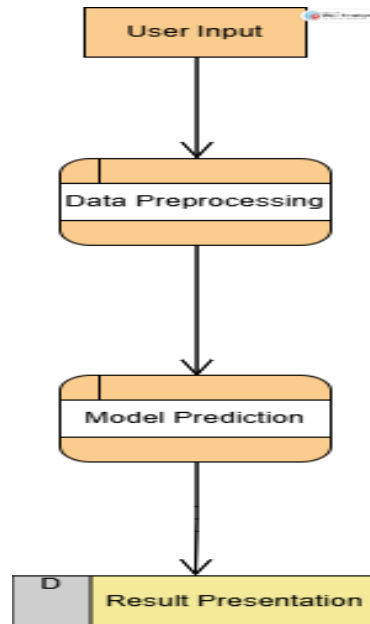
- **Accuracy:** The model should have high predictive accuracy.
- **User-friendliness:** Simple and easy-to-navigate interface.
- **Performance:** Model inference should be efficient and respond within seconds.

4. Project Design

4.1 Data Flow Diagrams

1. **User Inputs Data:** Users enter personal health metrics.
2. **Data Preprocessing:** Raw data is cleaned and normalized.
3. **Model Prediction:** The selected machine learning model predicts the likelihood of heart disease.

4. **Results Presentation:** The application displays the prediction result with a risk percentage and visualization.



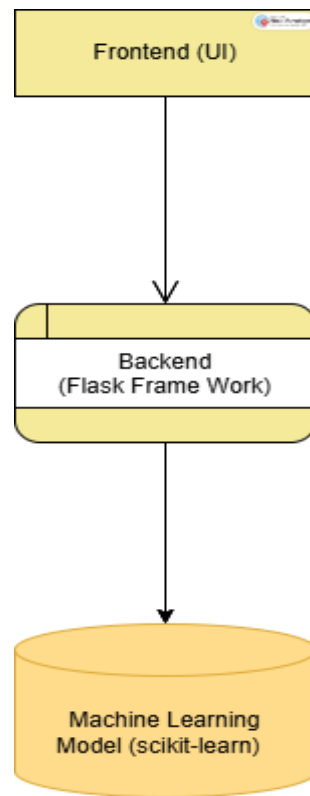
4.2 Solution Architecture

The architecture includes:

Frontend: HTML/CSS for UI, with JavaScript for form validation.

Backend: Flask framework to manage routes and prediction requests.

Machine Learning Model: Pre-trained model using Python's scikit-learn library, integrated for real-time predictions.



5. Coding & Solutioning

5.1 Feature 1: Data Preprocessing and Model Training

- **Data Cleaning:** Removed any inconsistent entries, handled missing values.
- **Data Encoding:** Converted categorical data (e.g., gender) into numerical values for model compatibility.
- **Scaling:** Used Standard Scaler to normalize features like age, cholesterol, and resting blood pressure.

Steps	Description
Label Encoding	Converted categorical data to numeric
Standard Scaling	Scaled data for model input
Train-Test Split	Split data 75% for training, 25% for testing

5.2 Feature 2: Web Application Interface and Prediction Functionality

- **Input Form:** A user-friendly form for entering data, with form validation.
- **Prediction Display:** Displays prediction results with a risk percentage and pie chart.

6. Performance Testing

6.1 Performance Metrics and Model Evaluation

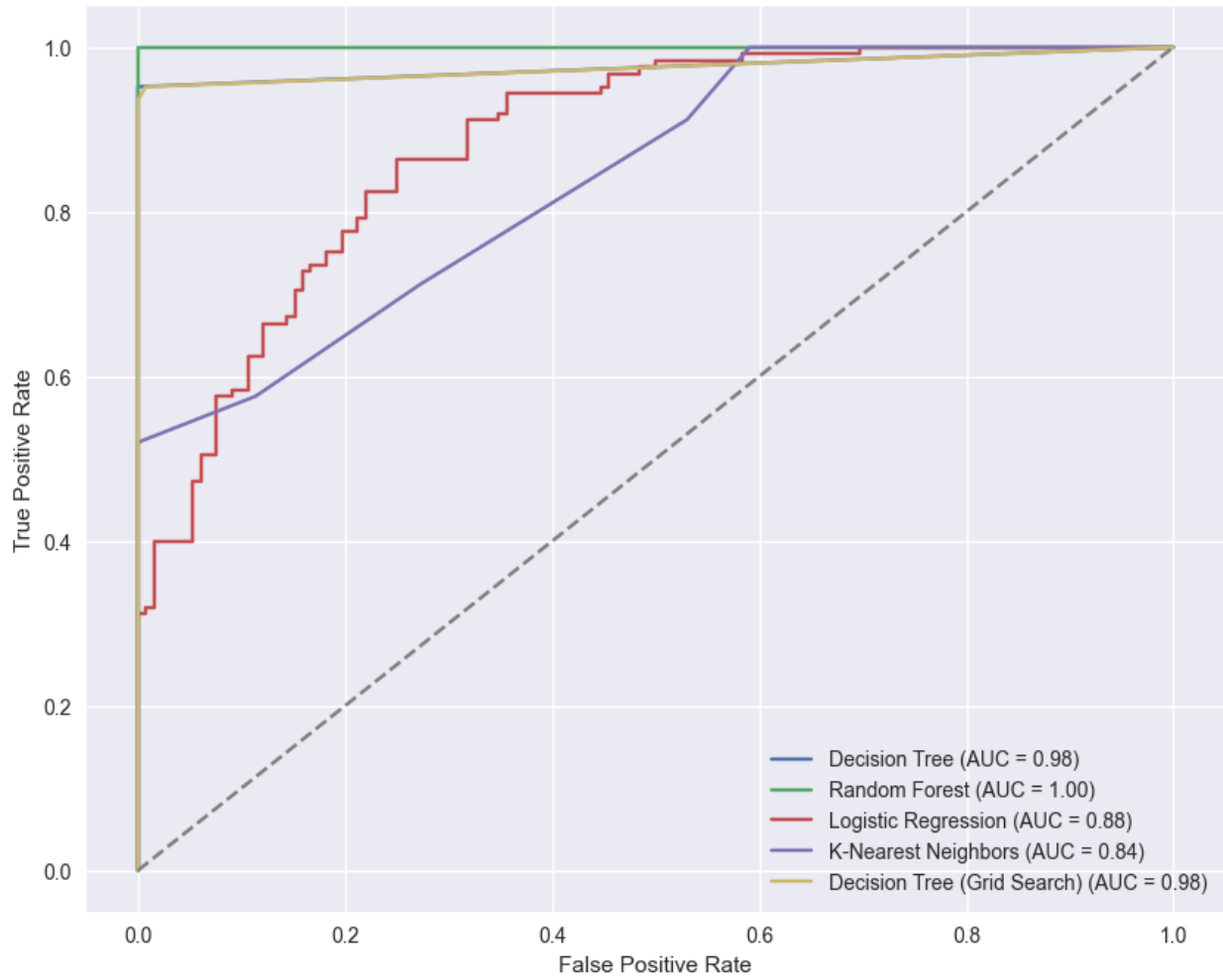
- **Models Tested:** Decision Tree, Random Forest, Logistic Regression, K-Nearest Neighbors.
- **Evaluation Metrics:** Used accuracy, precision, recall, F1-score, and AUC for each model.

Table: Model Performance Comparison

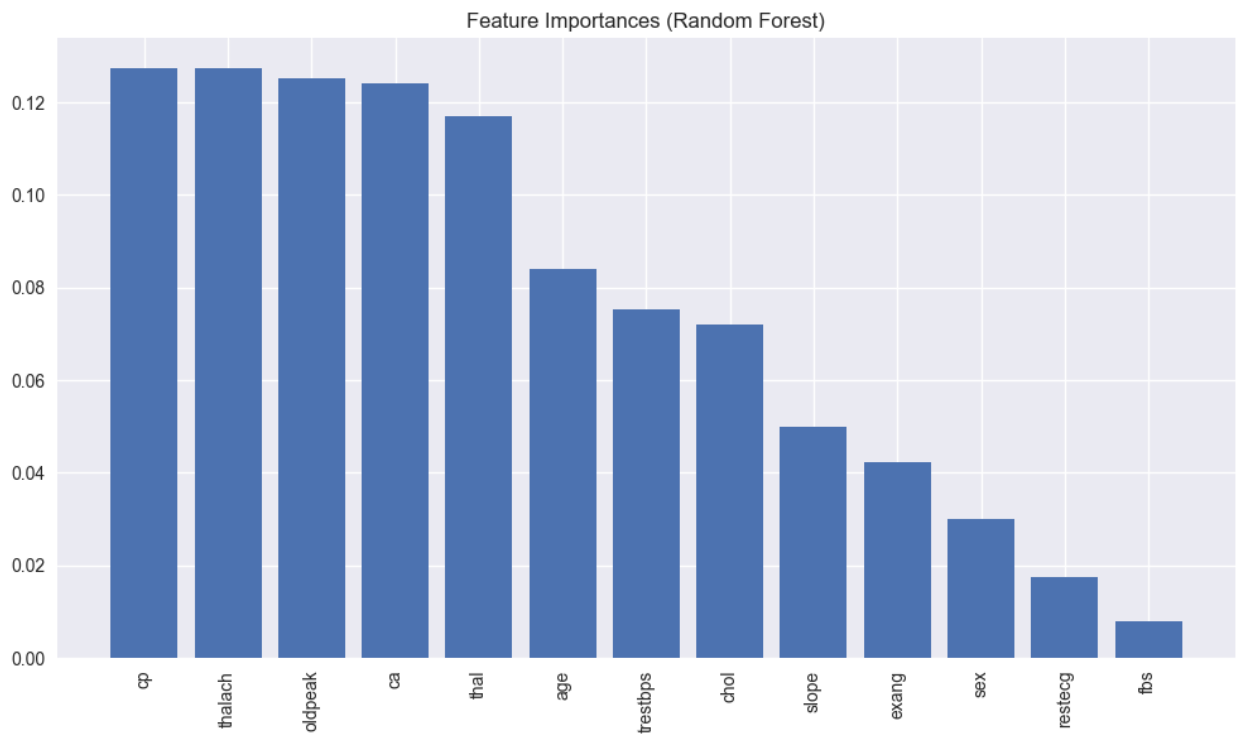
Model	Accuracy	Precision	Recall	F1 Score	AUC
Decision Tree	97.66%	100%	95.2%	97.5%	0.976
Random Forest	100%	100%	100%	100%	1.0
Logistic Regression	80.54%	76.6%	86.4%	81.2%	0.883
K-Nearest Neighbors	71.98%	71.2%	71.2%	71.2%	0.841

Graph: ROC Curves for Model Performance

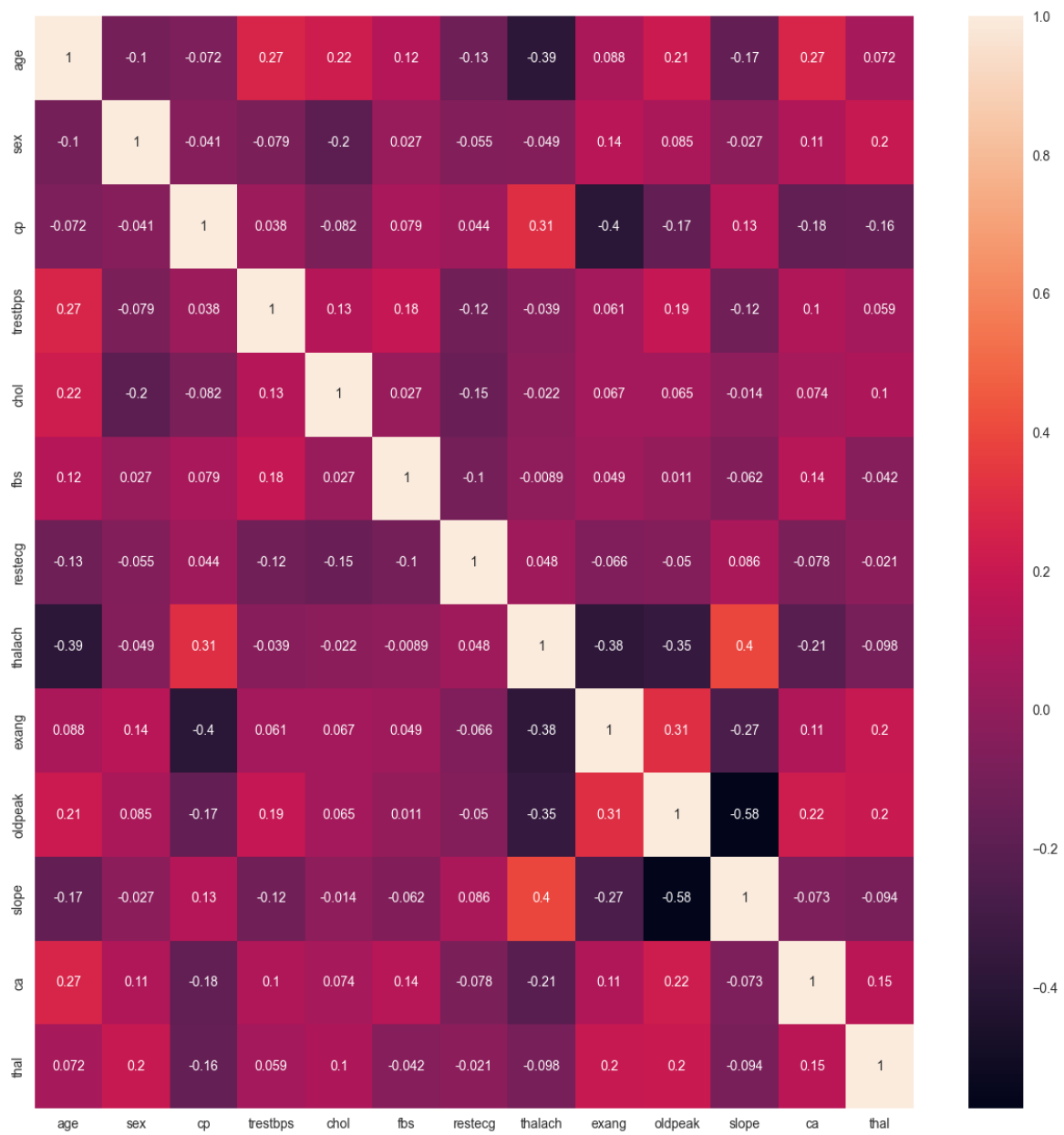
ROC Curves for Different Machine Learning Algorithms



Feature Importance

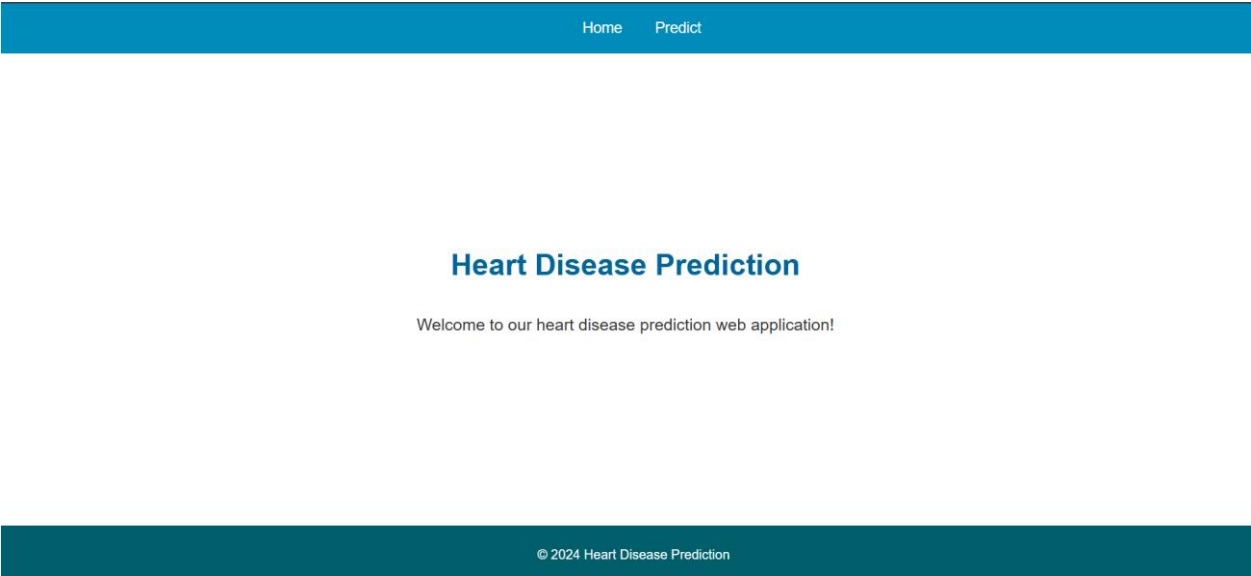


Correlation Matrix

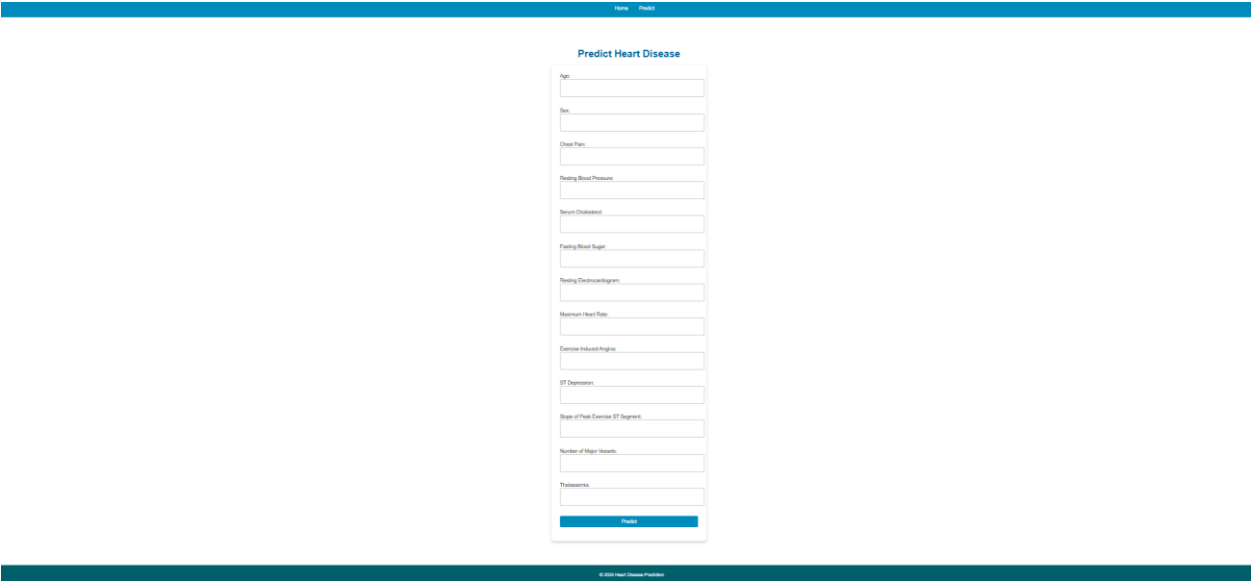


7. Results

7.1 Output Screenshots



Prediction Input Form



Results Page

Heart Disease Prediction Result



8. Conclusion

The heart disease prediction project demonstrated that machine learning could offer reliable early diagnosis tools. Random Forest was the best-performing model, achieving 100% accuracy, suggesting high reliability for predictive diagnostics.

9. Future Scope

Future improvements include:

- Incorporating larger and more diverse datasets for better generalizability.
- Exploring deep learning methods for potential accuracy improvements.
- Integrating with health information systems for real-world deployment.

10. Appendix

10.1 Source Code

[Link to source code.](#)

10.2 GitHub & Project Demo Link

[GitHub link.](#)